The Informational Value of Lobbying in the Tariff Exclusion Process*

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Abstract

This paper studies the informational role of lobbying in the U.S.–China trade war. Under Section 301 of the Trade Act of 1974, the United States imposed tariffs on more than $450 billion of Chinese imports and allowed firms to request product-specific exclusions. I develop a signaling model in which firms lobby to transmit information about the welfare consequences of exclusions to a politically motivated regulator who sets both the tariff and the approval rule. Calibrated to 50,000 exclusion requests linked to corporate-group lobbying disclosures, the model shows that informational lobbying moderated the Section 301 tariff and improved the targeting of tariff relief. As a result, total welfare losses fell by roughly 18 percent relative to a world without lobbying, even though the policymaker maximized a political rather than social objective. The framework provides a tractable approach for evaluating how bureaucratic discretion and private information interact in modern trade and industrial policy.

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1 Introduction

In 2018, the United States imposed a sweeping set of tariffs on imports from China under Section 301 of the Trade Act of 1974, citing unfair trade practices and intellectual-property violations. These measures eventually covered more than $450 billion in imports across four successive product lists. As documented by Bown and Kolb (2021), the tariffs disproportionately targeted intermediate inputs rather than final consumer goods, with rates ranging from 7.5 to 25 percent depending on the item. As part of the implementation, the Office of the U.S. Trade Representative (USTR) established a formal process through which firms could request product-specific exclusions from the new duties. Between 2018 and 2020, more than 50,000 petitions were submitted, each requiring the applicant to provide detailed information on sourcing patterns, domestic alternatives, and the expected economic harm from the tariff. The Government Accountability Office later documented the extraordinary informational burden this system placed on a small bureaucracy operating under tight deadlines and limited staff capacity. Reviewers were required to assess tens of thousands of product-level claims, often relying entirely on information self-reported by petitioners.

This setting, in which a politically directed policy must be implemented by a bureaucracy operating with limited information, raises a broader question of how lobbying affects policy through the transmission of information. The objective of this paper is to quantify the informational role of lobbying in the Section 301 tariff-exclusion process and to evaluate how this informational channel interacts with the political motives embedded in the government's objective, as well as the resulting implications for welfare. In the Section 301 context, these political motives are best interpreted as reduced-form weights that capture the distributional and strategic considerations surrounding the U.S.–China trade action, such as the importance assigned to domestic upstream producers, downstream users of Chinese inputs, tariff revenue, and exposure to Chinese suppliers. I show that lobbying improved the targeting of tariff relief, moderated the tariff set ex ante, and reduced the total welfare loss from the trade war by roughly 18 percent, even though the policymaker maximized a political rather than a welfare objective.

The Section 301 exclusion process offers a rare empirical window into the mechanics of informational lobbying. The introduction of large and heterogeneous tariffs triggered substantial lobbying activity. As documented by Chor et al. (2025), approval outcomes were systematically related to economic fundamentals such as reliance on Chinese suppliers. Building on this evidence, incorporating firm-level lobbying disclosures shows that lobbying did not guarantee success: many lobbying firms saw petitions denied, while some non-lobbying firms still received approvals. Taken together, these empirical patterns indicate that approval decisions reflected both publicly observable economic fundamentals and additional information supplied by firms. This perspective is consistent with the idea that lobbying in this setting operated as a costly communication channel through which firms conveyed credible private information about the economic consequences of tariffs, and that the USTR used this information, imperfectly but meaningfully, when deciding which products to exempt.

The institutional details reinforce this interpretation. The exclusion process unfolded within a rule-governed bureaucracy rather than a legislative or campaign-finance setting, and decisions were made within established administrative procedures. Firms interacted with the agency primarily by supplying technical and product-specific information, which case reviewers relied on heavily given tight time and staffing constraints. The USTR coordinated with multiple agencies (including U.S. Customs and the International Trade Commission) through a quasi-adjudicatory review process that emphasized documentation and administrability. In this environment, statutory criteria, administrative discretion, and limited information combined to place a premium on credible communication, making the setting naturally suited to an informational role for lobbying.

Motivated by these patterns, the paper develops a signaling model in which lobbying is interpreted as the costly transmission of private information about the economic consequences of tariff relief under a government objective that places reduced-form political weights on different components of welfare. These political weights summarize broader distributional and strategic considerations surrounding the Section 301 action, rather than being shaped by the firm–bureaucracy lobbying studied here. Firms privately observe the product-specific economic impact of an exclusion, information they know more precisely than the regulator, and decide whether to incur a fixed cost to lobby and thereby generate a noisy signal that the government interprets through Bayesian updating. The policy process unfolds in two stages: the government first chooses the Section 301 tariff while anticipating how firms will behave in the subsequent exclusion stage, and then, given realized lobbying decisions and signals, decides which petitions to approve. The signaling game is embedded in a multi-sector trade environment and evaluated under a politically weighted government objective that includes downstream surplus, upstream rents, tariff revenue, and the strategic importance of Chinese suppliers.

A key feature of the framework is that political considerations and informational transmission coexist. The government's objective contains reduced-form political weights that determine how it values domestic upstream producers, downstream firms, tariff revenue, and rents earned by Chinese suppliers. These weights reflect broader distributional and strategic priorities associated with the Section 301 action and are not shaped by the firm–bureaucracy lobbying studied here. Within this politically weighted environment, firms lobby to convey information about the part of the objective relevant to them. Importantly, the political weights do not change in the model; the role of lobbying is to make the government's politically constrained decisions better informed.

Three empirical facts motivate this structure. First, petitions submitted by firms that engaged in lobbying were more likely to be approved, even after controlling for industry, time, and sector characteristics. Second, consistent with Chor et al. (2025), approval outcomes varied systematically with economic indicators of domestic harm, such as reliance on Chinese inputs, indicating that the agency incorporated economically relevant information when evaluating petitions. Third, a substantial share of lobbying groups received no approvals across all their petitions, reflecting that lobbying was a costly activity with uncertain outcomes. These features, including costly participation, decisions that respond to observable fundamentals, and residual uncertainty, map directly into the signaling environment developed below.

I embed this signaling framework in a multi-sector quantitative environment calibrated to the universe of Section 301 exclusion petitions, combined with firm-level lobbying disclosures and sector-level trade data. Within each sector, products exhibit heterogeneous sourcing patterns based on pre–trade-war information, which generate variation in the product-specific economic impact of tariff relief. The model follows a three-stage sequence. First, the government chooses the Section 301 tariff while anticipating the informational environment. Second, firms privately observe the economic impact of an exclusion, summarized in the model as a type, and decide whether to incur the cost of lobbying in order to generate a signal. Third, the government evaluates petitions using the observed lobbying decisions and, when lobbying occurs, the noisy signals conveyed through those petitions, subject to idiosyncratic political shocks that capture residual randomness arising from factors such as reviewer assignment or transient shifts in political priorities. The calibration uses sector-level tariff outcomes, lobbying participation rates, and the approval rates of lobbying and non-lobbying petitions to identify the cost of lobbying, the informativeness of the signal, and the magnitude of these political shocks.

The calibrated parameters paint a clear picture. Two components of the government's reduced-form political weights are pinned down analytically: the MFN and Column 2 tariffs identify the weights on domestic upstream rents and downstream surplus. The remaining political weight, which governs how the government values Chinese upstream rents, is calibrated at the sector level by requiring the model's tariff-setting condition to match the observed Section 301 tariff. Lobbying signals are estimated to be highly informative relative to underlying sectoral heterogeneity, implying that the government learns effectively from lobbying when it occurs. The estimated idiosyncratic approval shocks are modest, which means outcomes were predictable on average but still retained the residual randomness inherent in the review process. The main friction lies in the fixed cost of participation, which deters many firms from lobbying. As a result, the information the government receives is reliable but limited in coverage, since only a subset of firms chooses to participate.

Quantitatively, the model shows that informational lobbying reduced the welfare cost of the tariffs by about 18 percent relative to a world without lobbying. The mechanism is twofold. First, lobbying lowers the average Section 301 tariff from 0.243 to 0.207. This moderation arises because the structural model captures the private cost of participation: the government internalizes that a higher tariff would induce more firms to incur lobbying costs, and lowers the rate to mitigate this response. Second, once the tariff is chosen, lobbying conveys precise signals that raise the approval rate from 4.8 to 7.9 percent, enabling more targeted relief for the highest-harm products. Crucially, the finding that these mechanisms improve aggregate welfare is a quantitative result rather than a theoretical necessity. In the calibrated environment, the information conveyed through lobbying allows the government to direct exclusions toward products for which tariff relief generates substantial efficiency gains, while the anticipatory tariff adjustment reduces the distortions created by the tariff itself. Together, these two forces generate a welfare improvement relative to a world without lobbying, even though the government maximizes a political rather than a welfare objective.

This paper contributes to four strands of literature. First, it builds on the literature on informational lobbying, which studies how lobbying conveys private information to policymakers (e.g., Potters and Van Winden, 1992; De Figueiredo and Silverman, 2008; Ludema et al., 2010). Second, it complements the large body of quid-pro-quo models in which lobbying purchases political influence (e.g., Grossman and Helpman, 1994; Goldberg and Maggi, 1999; Gawande et al., 2012), by focusing instead on settings where influence is already embedded in the policymaker's objective and lobbying primarily supplies information. Third, this paper is most closely related to the structural analysis of the Section 301 exclusion process by Chor et al. (2025). They document that approval outcomes correlate with economic fundamentals, consistent with information transmission, and quantify the welfare value of the exclusion mechanism itself, showing that it acts as a safety valve that allows the government to set higher tariffs. I extend their analysis by explicitly modeling the firm's endogenous decision to lobby, the precision of the resulting signals, and the political shocks governing approvals. This allows me to answer a distinct counterfactual question: rather than evaluating the exclusion process as a whole, I quantify the welfare effect of the lobbying channel, showing that the anticipation of costly signaling leads the government to moderate, rather than raise, the ex-ante tariff. Fourth, it connects to the literature on noisy signaling (e.g., Matthews and Mirman, 1983; Kartik, 2009), using these tools to characterize how information transmission operates within a politically weighted decision environment.

More broadly, the framework provides a template for analyzing administrative decision-making under incomplete information. Many modern industrial and trade policies such as Buy America waivers, export controls, investment screening, and environmental permitting require regulators to implement politically directed rules while relying on information provided by regulated parties. The Section 301 case shows that such systems can harness private information to improve implementation, even when the underlying objective is political rather than welfare maximizing.

The remainder of the paper proceeds as follows. Section 2 describes the institutional setting and exclusion process in detail. Section 3 presents empirical patterns motivating the signaling framework. Section 4 develops the theoretical model of informational lobbying. Section 5 calibrates the model to sector-level data and quantifies the informational value of lobbying. Section 6 presents counterfactual analyses that decompose the welfare effects of informational lobbying. Section 7 concludes.

2 Institutional Background

2.1 The Section 301 Exclusion Process

Section 301 of the Trade Act of 1974 authorizes the U.S. Trade Representative (USTR) to impose trade measures when foreign practices are deemed "unreasonable or discriminatory" and burden U.S. commerce. In August 2017, the USTR launched an investigation into China's technology-transfer, intellectual-property, and innovation policies. Following its findings in March 2018, the United States imposed four successive tariff lists, widely known as the Section 301 tariffs, covering more than $460 billion of annual imports from China. Each list was implemented through notice-and-comment procedures under the Administrative Procedure Act and published in the *Federal Register*. As coverage expanded from roughly $50 billion to about $460 billion, the measures ultimately encompassed almost 90 percent of U.S. imports from China, with tariff rates ranging between 7.5 and 25 percent.[1]

Recognizing that these tariffs could unintentionally harm U.S. producers and consumers dependent on Chinese inputs, the USTR established a formal exclusion process in July 2018.

[1] See Chor et al. (2025) for a detailed chronology of the four lists and their coverage.

This mechanism allowed importers, trade associations, and other stakeholders to request temporary, product-specific exemptions from the additional duties. Each notice of action opened a three-month filing window during which applicants were required to demonstrate that (i) the product was available only from China, (ii) the tariff would cause severe economic harm to U.S. interests, (iii) the product was not strategically linked to China's *Made in China 2025* program, (iv) granting the exclusion would not undermine the objectives of the investigation, and (v) that the product could be defined precisely enough for Customs to administer the exemption. These criteria guided reviewers but were not applied mechanically; no single factor was dispositive.[2] The third criterion, whether a product was associated with China's *Made in China 2025* industrial policy, reflected sector-specific strategic concerns and suggests that the government's sensitivity toward Chinese producers varied across industries.

Between 2018 and 2020, the USTR received roughly 53,000 exclusion petitions across the four tariff tranches. The agency initially relied on a small team of attorneys within its Office of General Counsel to review the first lists and later expanded capacity through detailees from other agencies, contract attorneys, and technical case experts as the volume of requests surged. The review followed a four-stage structure: (1) public submission and comment through an online portal, (2) internal substantive review by assigned analysts, (3) an inter-agency "administrability" assessment involving U.S. Customs, the International Trade Commission, and representatives from 22 federal agencies, and (4) publication of final decisions in the *Federal Register*. Approved exclusions were applied retroactively to the date of the original tariff and were product-based rather than firm-specific, meaning any importer of the qualifying good could claim relief. There was no formal appeal procedure, so petitioners had a single opportunity to make their case.

Administrative strain was a defining feature of the process. Average processing times reached 143 days for List 3 alone. Nearly 87 percent of petitions were denied, most for failing to demonstrate severe economic harm or to show lack of alternative supply. The GAO attributed these patterns to the USTR's limited staffing, absence of standardized guidance, and reliance on self-reported data from firms.[3] Reviewers therefore depended heavily on the credibility and clarity of information voluntarily supplied by petitioners. This delegation of information provision effectively turned the exclusion process into an informal communication channel between private actors and a resource-constrained bureaucracy.

By early 2020, the share of U.S. imports from China subject to Section 301 tariffs exceeded

[2] Government Accountability Office (2021).
[3] Government Accountability Office (2021), pp. 2, 10–14.

50 percent, while products excluded from tariffs accounted for about 10 percent before the program wound down. The combination of massive petition volume, multi-agency review, and limited administrative capacity made the exclusion process one of the largest quasi-judicial exercises in modern U.S. trade policy, offering a natural setting for studying how lobbying and information transmission shape bureaucratic decision-making under political constraints.

The combination of open participation and limited administrative capacity created fertile ground for lobbying activity. Unlike legislative lobbying, which targets elected officials and emphasizes access, persuasion, and influence, exclusion-related lobbying targeted bureaucrats tasked with quasi-adjudicatory decisions. Firms could not alter statutory criteria but could shape the *information* available to reviewers, for example, by providing technical evidence on input substitution or production bottlenecks.

In parallel with the USTR's formal submission portal, firms engaged in lobbying activities governed by the Lobbying Disclosure Act (LDA). Disclosures compiled by the Center for Responsive Politics and Kim (2018) show a sharp rise in filings mentioning "Section 301 tariffs" or relevant issues beginning in 2018, documenting several hundred firms and trade associations that hired lobbyists to represent them or self-represented before the USTR. This registered lobbying provides a credible, legally-monitored record of firms' engagement.

3 Empirical Motivation

This section documents the empirical patterns that motivate the theoretical framework. The goal is twofold: first, to show that approval decisions were systematically linked to both lobbying participation and indicators of economic harm (Section 3.2); and second, to show that approval outcomes exhibited uncertainty, with patterns consistent with an informational channel in which lobbying conveys useful but imperfect information (Section 3.3). Together, these findings motivate a model in which lobbying operates as a costly mechanism for transmitting information to the regulator.

3.1 Overview of Section 301 Exclusion Petitions

Table 1 provides a summary of the petition universe. Across all four tranches, roughly nine percent of applicant firms engaged in lobbying that explicitly referenced Section 301 in their disclosure filings. Approval rates declined sharply over time, from more than 30 percent in the first two tranches to below 7 percent in the final lists, consistent with the rise in workload and tighter political oversight as coverage expanded.

Table 1: Overview of Section 301 Exclusion Petitions

Tranche	Total Applications	Unique Companies	Lobbied (%)	Approval Rate (%)	Approval (Lobbied, %)
Overall	52,745	4,575	9.17	12.90	22.38
Tranche 1	10,813	1,200	12.59	32.81	33.11
Tranche 2	2,869	453	13.38	37.43	62.76
Tranche 3	30,283	2,513	7.45	4.94	14.13
Tranche 4	8,780	1,249	9.53	6.55	10.27

Notes: Each row reports the total number of exclusion petitions and approval outcomes by tariff tranche.

These aggregates already suggest two salient features. First, lobbying participation was selective: only a small fraction of petitioners invested in formal lobbying, suggesting that the cost of participation was substantial relative to the associated increase in approval probability. Second, while lobbying correlates positively with approval rates, the relationship is far from mechanical: many lobbying firms still saw all their petitions denied. This pattern is consistent with a setting in which lobbying is costly and informative and in which approval outcomes remain uncertain rather than guaranteed.

The remainder of this section explores the underlying structure of these patterns.

3.2 Determinants of Exclusion Approval

Data sources. The empirical analysis links exclusion petitions from the USTR public docket to firm-level lobbying disclosures from LobbyView, consolidating subsidiaries under common corporate-group identifiers from the Bureau van Dijk *Orbis* database. This aggregation is essential, as firms often channel lobbying expenditures through a parent entity while submitting exclusion petitions through multiple operational affiliates. Industry characteristics and trade elasticities are drawn from Soderbery (2015), while tariff schedules come from the U.S. International Trade Commission (USITC) and import values from the U.S. Census Bureau. To construct product-level measures of domestic reliance, I combine data from the 2017 BEA Input–Output tables and the 2017 Economic Census Product Statistics. I start from sector-level domestic production totals in the BEA tables and disaggregate them to the product level using shipment-value weights derived from the Economic Census. This approach allocates each sector's domestic production across products within the same *North American Industry Classification System* (NAICS) industry, yielding granular domestic-to-import ratios for the *Harmonized Tariff Schedule* (HTS)–NAICS concordance used in the analysis. The resulting dataset covers the full population of Section 301 petitions and associated lobbying activities at the product level, mapped to input–output sectors used later in the calibration.

Specification. The baseline regression takes the exclusion-approval indicator as the dependent variable. The key explanatory variable is a dummy for whether the firm group (`guo25`) engaged in lobbying that explicitly targeted Section 301 in the same tranche. Control variables include the standardized import demand elasticity (σ), the standardized inverse export supply elasticity (ω), the China import share at the HTS 10-digit level, the domestic share constructed from the BEA Input–Output and Economic Census data described above, and the number of exclusion applications filed under the same HTS product code. The specification includes tranche, year, and industry fixed effects. I estimate the model using both an ordinary least squares linear probability model (LPM) and a logit specification, as reported in Table 2.

Table 2: Determinants of Exclusion Approval

	(1) OLS (LPM)	(2) Logit
Lobbied	0.0500^{**}	0.376^{**}
	(0.0238)	(0.191)
Demand Elasticity (σ)	-0.0156^{***}	-0.156^{***}
	(0.0021)	(0.0347)
Inverse Supply Elasticity (ω)	-0.0029^{***}	-0.148
	(0.0009)	(0.112)
China Import Share	0.309^{*}	3.343^{**}
	(0.185)	(1.603)
Domestic Share	0.265^{*}	2.288^{*}
	(0.145)	(1.194)
Tranche & Year FEs	Yes	Yes
Industry FE	Yes	Yes
Observations	52,355	52,355

Notes: Dependent variable = indicator for exclusion approval. All specifications include tranche, year, and industry fixed effects. Each specification also controls for the number of exclusion applications per HTS product code (coefficients not reported). Standard errors are two-way clustered by corporate group (`guo25`) and six-digit HTS code. Elasticities (σ and ω) are standardized to mean 0 and unit variance.

Results and interpretation. Lobbying is a strong positive predictor of approval: petitions associated with lobbying firms are about 5.0 percentage points more likely to be granted in the OLS model, and the implied marginal effect from the logit specification is approximately 3.4 percentage points (average marginal effect). Given an unconditional approval rate of 13 percent, this represents a sizable difference even after accounting for industry and time fixed effects.

Beyond lobbying, approval probabilities vary systematically with sectoral fundamentals in ways consistent with the USTR's stated goal of mitigating domestic harm. The economic interpretation of the coefficients is straightforward.

- A higher **import demand elasticity** ($\boldsymbol{\sigma}$) means that buyers can more easily substitute away from affected suppliers. Greater substitutability allows firms to avoid tariff burdens, resulting in lower domestic harm and, consequently, a lower likelihood of approval.
- A higher **inverse export supply elasticity** ($\boldsymbol{\omega}$) implies that foreign suppliers have more inelastic supply, causing them to absorb a larger share of the tariff through lower export prices. This reduction in domestic hardship *should* correspond to a lower probability of approval. We find a significant negative relationship in the OLS model (Col. 1), though the point estimate in the logit specification (Col. 2), while negative, is not statistically precise.
- Finally, products with higher **China import shares** or higher **domestic shares** are significantly more likely to be approved, reflecting the government's responsiveness to sectors more directly exposed to tariff-induced disruption.

Together, these patterns indicate that the agency's decisions aligned with economic harm considerations. The regression results therefore support two complementary findings: lobbying independently predicts approval outcomes, and the pattern of coefficients on σ, ω, and exposure variables is consistent with the government acting on economically relevant information. At the same time, lobbying did not guarantee success, and non-lobbying firms also secured approvals, indicating residual variation beyond observable sectoral characteristics.

3.3 Uncertainty and Imperfect Screening

The data indicate that the relationship between lobbying and approval outcomes is noisy and imperfect. Table 3 summarizes success rates at the *corporate-group* level. A substantial share of lobbying groups had every petition denied, despite in some cases having filed a large number of requests. Conversely, aggregate data show that roughly 9 percent of petitions submitted by non-lobbying firms were approved.

These patterns motivate one core feature of the theoretical framework. In the data, both lobbying and non-lobbying firms receive a mix of approvals and denials, indicating that approval was not mechanically determined by participation status. To capture this, the model incorporates *idiosyncratic political shocks* that introduce residual uncertainty into the approval decision.

In addition, while lobbying is associated with higher approval rates, it does not transmit information perfectly. Submissions vary in clarity, and agencies do not always interpret complex materials with full precision. To reflect this imperfect screening, the model allows for *signal noise*. This feature makes the informational channel more realistic and gives the framework enough flexibility to accommodate the magnitude of the approval-rate gap between lobbying and non-lobbying petitions.

Table 3: Lobbying Corporate Groups with Zero Approvals

Metric	Zero-Approval Groups	All Lobbying Groups
Number of Groups	89	203
Total Petitions	1,310	4,839
Mean Petitions per Group	14.7	23.8
Median	4	5
Maximum	201	655
Minimum	1	1

Notes: Each observation represents a corporate group (`guo25`) aggregating all affiliated firms' petitions. Zero-approval groups submitted at least one petition but received no approvals.

3.4 Summary

Three empirical regularities emerge from the Section 301 data:

1. **Lobbying predicts approval.** Petitions associated with lobbying firms are substantially more likely to be granted, even after controlling for industry, time, and sectoral characteristics (Section 3.2).

2. **Approvals track economic harm.** The probability of approval varies systematically with economic fundamentals such as substitutability and import exposure, suggesting that the regulator incorporated economically relevant information when evaluating petitions (Section 3.2).

3. **Approval is uncertain for all firms.** Some lobbying groups received no approvals despite extensive participation, and a nontrivial share of non-lobbying firms secured approvals (Section 3.3). These features indicate that approval was not mechanically tied to participation and that success remained uncertain.

Taken together, these patterns suggest a consistent interpretation. Consistent with Chor et al. (2025), approval outcomes correlated with indicators of economic harm, which points to a decision process that responded to information about potential domestic damage. The patterns documented here complement their evidence by showing that this informational transmission was imperfect. While lobbying is strongly associated with a higher probability of success, the data also show substantial residual uncertainty, including lobbying denials and non-lobbying approvals, which indicates that the process was noisy rather than deterministic. These empirical regularities motivate the theoretical framework developed next, which relaxes the full-information structure in prior work and models lobbying as a costly and noisy signaling mechanism.

4 Theoretical Framework

4.1 Overview

This section develops a model in which lobbying serves as a costly signal that conveys private information about how an exclusion affects the government's politically weighted objective. The framework embeds a noisy signaling game between firms and the government, where firms privately observe the economic component of this objective and decide whether to incur a cost to reveal it. The government interprets lobbying through Bayesian updating and makes approval decisions subject to political uncertainty. The resulting structure delivers a tractable mapping, through a cutoff lobbying strategy and a posterior-based approval rule, between lobbying participation, approval probabilities, and the underlying impact of exclusions under the government's objective. The signaling structure analyzed in this section constitutes the second stage of a broader policy environment. In the full model the government chooses the Section 301 tariff in an earlier stage, anticipating the exclusion game characterized below; the signaling analysis therefore proceeds conditional on that tariff choice.

4.2 Economic Environment and Timing

Consider a continuum of goods $g \in [0, 1]$ within a sector. For each good, a downstream producer, representing the firm that petitions for an exclusion, decides whether to lobby, and the government determines whether to grant the exemption. The timing is as follows:

1. The firm privately observes its type θ_g, the *economic component* of the change in the government's politically weighted objective from granting an exclusion.

2. The firm chooses whether to lobby ($d_g = 1$) or not ($d_g = 0$). Lobbying requires a fixed cost $c > 0$, and the government observes the firm's lobbying decision.

3. If the firm lobbies, it generates a noisy signal

$$s_g = \theta_g + \varepsilon_g, \qquad \varepsilon_g \sim \mathcal{N}(0, \sigma_\varepsilon^2),$$

which the government observes. No signal is received when $d_g = 0$.

4. Given its information set, consisting of the observed lobbying decision and, if applicable, the signal, the government forms a posterior belief about θ_g and applies an approval rule. The final decision reflects an idiosyncratic political shock $\xi_g \sim \mathcal{N}(0, \eta^2)$, which shifts the approval threshold but does not alter the underlying objective impact.

In this environment, θ_g represents the part of the government's politically weighted objective that is known to the firm and must be inferred by the government. The political shock ξ_g introduces randomness into approval outcomes, ensuring that decisions are not perfectly predictable even when information is precise, while leaving the underlying objective impact unchanged.

4.3 Information and Beliefs

All parameters of the exclusion stage, including the approval-rule shock variance η^2 and the mapping from expected objective impact to approval probabilities, are common knowledge. The only asymmetric information at this stage concerns the realization of the firm-specific component θ_g, which summarizes the economic impact of an exclusion under the government's objective and is observed by the firm but not by the government.

Information structure without parametric priors. We do not impose a parametric distribution on types θ_g. Let $\pi(\theta)$ denote the economy-wide density of firm types. In equilibrium, firms with $\theta_g \geq \underline{\theta}$ choose to lobby, so the government's prior over the types of lobbyists is the truncated density

$$\pi_\ell(\theta) = \frac{\pi(\theta)\,\mathbf{1}\{\theta \geq \underline{\theta}\}}{\Pr(\theta \geq \underline{\theta})}.$$

The government observes whether the firm lobbies, and when $d_g = 1$, it also observes a noisy signal

$$s = \theta + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2).$$

Let $f_S(s)$ denote the marginal density of signals implied by $\pi_\ell(\theta)$ and the Gaussian noise. The posterior mean for a lobbying firm satisfies *Tweedie's formula*:

$$\mathbb{E}[\theta \mid s,\ \theta \geq \underline{\theta}] \;=\; s \;+\; \sigma_\varepsilon^2 \, \frac{\mathrm{d}}{\mathrm{d}s} \log f_S(s). \tag{1}$$

This expression holds for any prior density and relies only on normal signal noise; the effect of the truncation operates through the shape of $f_S(s)$.

Because the signal is Gaussian with additive noise, the likelihood $f(s \mid \theta) = \phi(s - \theta)$ satisfies the monotone likelihood ratio property, so higher types generate stochastically higher signals. By the standard "good news" result (Milgrom 1981), this implies that the posterior distribution $\pi(\theta \mid s)$ is increasing in s in the sense of first-order stochastic dominance for any prior, including the truncated prior $\pi_\ell(\theta)$. Since the government's approval rule is increasing in the posterior mean, higher true types place more weight on high-signal realizations and therefore receive higher approval probabilities in expectation.

How noise shapes approval when lobbying. For a lobbying firm of type θ, the approval probability integrates over signal realizations:

$$P_1(\theta; \sigma_\varepsilon, \eta, \underline{\theta}) = \mathbb{E}_\varepsilon\left[\Phi\left(\tfrac{\mathbb{E}[\theta|s,\theta \geq \underline{\theta}]}{\eta}\right)\right], \qquad s = \theta + \varepsilon.$$

Noise σ_ε affects $P_1(\theta)$ through two distinct channels:

1. *Posterior compression.* From (1), a larger σ_ε dampens curvature in $\log f_S(s)$, reducing the sensitivity of the posterior mean to the signal. In the limit $\sigma_\varepsilon \to \infty$, the posterior collapses to the mean of the lobbying pool, making $P_1(\theta)$ nearly constant across types.
2. *Signal dispersion.* A larger σ_ε spreads the distribution of s around θ, altering the probability that posterior beliefs fall above the implicit approval threshold.

Because the lobbying pool is truncated from below, its mean exceeds the type of the marginal lobbyist. Posterior compression therefore tends to raise P_1 for marginal types and lower it for high types; the net effect is determined empirically.

Cost and decision shock. The lobbying cost c determines the participation cutoff $\underline{\theta}$: a higher cost leads to a higher cutoff and fewer lobbyists. The political shock η enters the government's approval rule and increases randomness in outcomes. Larger η makes approvals less responsive to the expected objective impact, narrowing the difference between P_1 and P_0, with both probabilities evaluated using the appropriate posterior beliefs.

4.4 Firm Behavior

Each firm derives a private benefit $B(\theta_g)$ if the exclusion is granted, where $B'(\theta_g) > 0$.[4]

For a given cutoff $\underline{\theta}$ and common parameters $(\sigma_\varepsilon, \eta)$, let

$$P_1(\theta_g; \underline{\theta})$$

denote the approval probability for a lobbying firm of type θ_g, and

$$P_0(\underline{\theta})$$

the approval probability for a non-lobbying firm (which depends on the cutoff only through the truncated prior). The expected payoff from lobbying ($d_g = 1$) or not lobbying ($d_g = 0$) is then

$$U_1(\theta_g; \underline{\theta}) = B(\theta_g)\, P_1(\theta_g; \underline{\theta}) - c, \tag{2}$$

$$U_0(\theta_g; \underline{\theta}) = B(\theta_g)\, P_0(\underline{\theta}). \tag{3}$$

The firm lobbies if and only if $U_1(\theta_g; \underline{\theta}) \geq U_0(\theta_g; \underline{\theta})$. Under Gaussian additive noise, posterior beliefs are increasing in the signal, and the approval rule is increasing in these posteriors. These properties imply that best responses are monotone in type, so equilibrium lobbying decisions take the form of a cutoff rule:

$$d_g = \begin{cases} 1 & \text{if } \theta_g \geq \underline{\theta}, \\ 0 & \text{if } \theta_g < \underline{\theta}, \end{cases}$$

for some cutoff $\underline{\theta}$.

The marginal type $\underline{\theta}$ is indifferent, satisfying

$$B(\underline{\theta})\big[P_1(\underline{\theta}; \underline{\theta}) - P_0(\underline{\theta})\big] = c. \tag{4}$$

The left-hand side of (4) is continuous in $\underline{\theta}$ because $B(\theta)$, $P_1(\theta; \underline{\theta})$, and $P_0(\underline{\theta})$ are continuous in type and in the cutoff. Therefore, for any $c > 0$, a cutoff $\underline{\theta}$ solving (4) exists, possibly at

[4] The condition $B'(\theta_g) > 0$ ensures that types with a larger positive impact on the government's objective also have more to gain privately from an exclusion. Section 4.8 shows how both θ_g and $B(\theta_g)$ arise from the same underlying primitives. Under a mild parameter restriction—derived in Appendix (deferred to future version)—the two objects inherit the same monotonicity, validating the sorting assumption used here.

a boundary (all firms lobby or none lobby).

4.5 Government Decision Rule

The government approves an exclusion whenever the expected change in its politically weighted objective, given the available information, exceeds the idiosyncratic political shock ξ_g:

$$\mathbb{E}[\theta_g \mid \text{Information}] + \xi_g \geq 0, \tag{5}$$

with $\xi_g \sim \mathcal{N}(0, \eta^2)$. Conditional on any posterior mean m, the approval probability is

$$\Phi\left(\frac{m}{\eta}\right),$$

so the only difference across firms arises from differences in posterior beliefs.

Lobbying firms. If a firm lobbies, the government observes the noisy signal $s_g = \theta_g + \varepsilon_g$ with $\varepsilon_g \sim \mathcal{N}(0, \sigma_\varepsilon^2)$. Given the lobbying prior truncated at $\underline{\theta}$, the posterior mean is

$$\mu_1(s_g) = \mathbb{E}[\theta_g \mid s_g,\, \theta_g \geq \underline{\theta}].$$

Integrating over the signal noise yields the approval probability for a lobbying type θ_g:

$$P_1(\theta_g; \underline{\theta}) = \mathbb{E}_{\varepsilon_g}\left[\Phi\left(\frac{\mu_1(s_g)}{\eta}\right)\right], \qquad s_g = \theta_g + \varepsilon_g. \tag{6}$$

Non-lobbying firms. If a firm does not lobby, the government receives no signal and relies on the truncated prior over non-lobbying types. Let

$$\mu_0 = \mathbb{E}[\theta_g \mid \theta_g < \underline{\theta}]$$

denote the mean of this prior. The resulting approval probability is

$$P_0(\underline{\theta}, \eta) = \Phi\left(\frac{\mu_0}{\eta}\right). \tag{7}$$

Notation. For readability, the main text writes $P_1(\theta_g)$ and P_0 compactly. Both are equilibrium objects that depend on the cutoff $\underline{\theta}$ through the truncated priors and posterior beliefs. When needed, we write the full dependence explicitly as $P_1(\theta_g; \underline{\theta}, \sigma_\varepsilon, \eta)$ and $P_0(\underline{\theta}, \eta)$.

4.6 Equilibrium Characterization

An equilibrium consists of a cutoff $\underline{\theta}$ and approval probabilities (P_1, P_0) such that:

1. Firms follow the cutoff rule implied by (4);
2. The government forms correct Bayesian beliefs given this strategy and the Gaussian signal structure;
3. Approval probabilities are consistent with the government's decision rule (5).

These conditions define a Perfect Bayesian Equilibrium (PBE) of the signaling game. Because higher types generate stochastically higher signals under Gaussian noise, and because the approval rule is increasing in posterior beliefs, higher types receive higher approval probabilities in expectation. Thus equilibrium behavior is separating: high-type firms lobby and obtain higher approval odds, while low-type firms abstain and face lower approval probabilities.

Comparative statics. Three parameters govern equilibrium outcomes: the lobbying cost c, the signal noise σ_ε, and the political shock η.

- *Lobbying cost c.* A higher cost raises the cutoff $\underline{\theta}$, reducing the share of firms that lobby.
- *Signal noise σ_ε.* Smaller noise makes the signal more informative, steepening the mapping from type to posterior mean and increasing the dispersion of approval probabilities among lobbyists. Larger noise compresses posteriors toward the lobbying-pool mean, flattening $P_1(\theta)$ and weakening type discrimination.
- *Political shock η.* A larger η introduces more randomness into the approval rule and makes decisions less responsive to expected objective impacts. This primarily affects the non-lobbying approval rate P_0 and narrows the difference between P_1 and P_0.

4.7 Connection to the Objective Component

In the signaling environment, a firm's private type θ_g summarizes the change in the government's politically weighted objective that would result from granting an exclusion. Lobbying decisions therefore provide information about this quantity: firms with higher θ_g have stronger incentives to lobby, and the government interprets lobbying and signal realizations as noisy indicators of the underlying objective impact. The signaling game thus operates on a reduced-form representation of the government's objective, with θ_g capturing the relevant economic component that is known to the firm but not to the government.

The next subsection microfounds θ_g by deriving the government's objective from a structural model of production, input demand, tariff incidence, and political weighting. This microfoundation delivers an explicit expression for the objective change associated with a tariff exclusion and links the reduced-form type θ_g to underlying technological and policy primitives.

Before introducing the microfoundations for θ_g, it is useful to note that the Section 301 tariff is chosen by the government in an earlier stage of the policy process. In the full structural environment, the government selects the initial tariff anticipating the equilibrium of the exclusion stage described above. The signaling model therefore characterizes the second stage of a larger game, conditional on the tariff chosen in the first stage. Once the objective is microfounded below, this structure allows the model to map the tariff choice into the distribution of economic types $\{\theta_g\}$ that governs lobbying behavior and exclusion decisions.

4.8 Microfounding the Objective Component

Before introducing the structural environment, it is helpful to clarify the information structure. All sectoral primitives that enter the government's objective—such as demand and supply elasticities, technology parameters, political weights, and the approval-rule shock variance η^2—are common knowledge. The only asymmetric information concerns the firm-specific component θ_g, which reflects the change in the government's objective from granting an exclusion. In the structural model that follows, θ_g is a known function of these common sectoral primitives and of good-specific supply shifters that are observed by firms but not by the government. Heterogeneity in θ_g across goods arises from these idiosyncratic shifters; conditional on them, the mapping from primitives to θ_g is common knowledge. The remainder of this subsection derives θ_g from the underlying technological and policy primitives.

Each sector–good pair (s, g) uses intermediate inputs supplied by domestic, Chinese, and rest-of-world producers to generate a final good. Because these final goods are traded in perfectly competitive world markets, their prices are fixed from the U.S. perspective. As a result, tariff exclusions do not affect aggregate consumer demand. All variation in the government's politically weighted objective therefore arises through changes in upstream rents, downstream surplus, and tariff revenue. By deriving downstream behavior from a profit-maximization problem—rather than from quasilinear utility—the analysis preserves the fixed-expenditure structure found in earlier work while grounding it in a production-side environment. The structural expression for the change in the objective when the Section 301 tariff is removed provides the counterpart of the reduced-form type θ_g.

Final Consumers and Aggregate Demand. To anchor downstream production in a demand system, each good (s,g) produces a final output Y_{sg} that enters a representative consumer's quasi-linear utility function together with a freely traded numeraire:

$$U = X_0 \; + \; \left[\int_s \int_{g \in s} \phi_{sg}^{\frac{1}{\rho}} Y_{sg}^{\frac{\rho-1}{\rho}} \, dg \, ds \right]^{\frac{\rho}{\rho-1}}, \qquad \rho > 1.$$

Because these final goods are traded in world markets under perfect competition, their prices are fixed at $P_{Y,sg} = 1$. Quasi-linearity implies that total expenditure on the CES aggregate is invariant to tariff policy, and the fixed world prices imply that the quantity demanded of each Y_{sg} does not change when a tariff exclusion is granted. Consequently, exclusions do not affect welfare through consumer demand or final-good prices. All policy-relevant variation instead enters through intermediate-input costs faced by downstream firms, which determine upstream producer surplus, downstream surplus, and tariff revenue. This ensures that the microfoundations developed below fully determine the structural counterpart of θ_g.

Downstream Technology and Input Demand. Each variety (s,g) is produced by a competitive downstream firm that combines a composite intermediate input X_{sg} into final output. Output is given by

$$Y_{sg} = B_s \ln X_{sg},$$

where $B_s > 0$ is a sectoral technology parameter. The logarithmic form is chosen because its first-order condition delivers a constant level of total expenditure on intermediate inputs. The composite input aggregates domestic (h), Chinese (c), and rest-of-world (f) varieties through a CES aggregator with elasticity of substitution $\sigma_s > 1$:

$$X_{sg}^{\frac{\sigma_s-1}{\sigma_s}} = x_{sgh}^{\frac{\sigma_s-1}{\sigma_s}} + x_{sgc}^{\frac{\sigma_s-1}{\sigma_s}} + x_{sgf}^{\frac{\sigma_s-1}{\sigma_s}}.$$

With the final-good price normalized to 1, the downstream producer chooses X_{sg} to maximize

$$\Pi_{sg}^{\text{down}} = B_s \ln X_{sg} - P_{sg} X_{sg},$$

where P_{sg} is the CES input price index. The first-order condition, $\frac{B_s}{X_{sg}} = P_{sg}$, implies

$$P_{sg} X_{sg} = B_s.$$

Thus total spending on intermediate inputs,

$$M_{sg} \equiv P_{sg}X_{sg},$$

is constant and equal to B_s. This property ensures that tariff exclusions affect downstream firms only through input-price changes, not through changes in the scale of production.

Upstream Supply. Each origin $i \in \{h, c, f\}$ supplies an intermediate-input variety for (s, g) using labor L_{sgi} and a specific factor K_{sgi}. Production follows the standard specific-factor technology

$$q_{sgi} = A_{sgi}\, L_{sgi}^{\frac{1}{1+\omega_s}}\, K_{sgi}^{\frac{\omega_s}{1+\omega_s}}, \qquad \omega_s > 0,$$

and firms are perfectly competitive, taking factor prices as given with the wage normalized to 1. Solving the firm's profit-maximization problem yields optimal labor demand

$$L_{sgi} = \left(\frac{p_{sgi}A_{sgi}}{1+\omega_s}\right)^{\frac{1+\omega_s}{\omega_s}} K_{sgi},$$

which implies the supply function

$$q_{sgi} = a_{sgi}^{-1/\omega_s}\, p_{sgi}^{1/\omega_s}, \qquad a_{sgi} \equiv (1+\omega_s)\, K_{sgi}^{-\omega_s}\, A_{sgi}^{-(1+\omega_s)}.$$

With an ad valorem tariff τ_{sgi}, downstream expenditure on origin i is $(1 + \tau_{sgi})p_{sgi}q_{sgi}$, and using $M_{sg} = P_{sg}X_{sg}$, the corresponding expenditure share is

$$s_{sgi} = (1+\tau_{sgi})\frac{a_{sgi}^{-1/\omega_s}}{M_{sg}} p_{sgi}^{(1+\omega_s)/\omega_s}.$$

Because the specific factor earns rents while labor is paid its marginal product, upstream operating profits are a constant fraction of revenue:

$$\Pi_{sgi}^{\text{up}} = \frac{\omega_s}{1+\omega_s}\, p_{sgi}q_{sgi} = \frac{\omega_s}{1+\omega_s}\frac{M_{sg}s_{sgi}}{1+\tau_{sgi}}.$$

Downstream Surplus and the Link to Shares. Given the downstream firm's optimal choice $P_{sg}X_{sg} = B_s$, the indirect profit function depends only on the CES input price index:

$$\Pi_{sg}^{*}(P_{sg}) = B_s \ln\!\left(\frac{B_s}{P_{sg}}\right) - B_s.$$

The policy-relevant component is therefore

$$\Pi_{sg}^{\text{down, policy}} = -B_s \ln P_{sg}.$$

Combining the CES aggregator with upstream supply yields the share–price-index relationship

$$\ln P_{sg} = \kappa_s \ln s_{sgh} \ + \ \text{constant}, \qquad \kappa_s \equiv \frac{\omega_s \sigma_s + 1}{(1+\omega_s)(\sigma_s - 1)}.$$

The constant term collects components unaffected by the Chinese tariff and does not influence policy comparisons. Substituting this expression yields the downstream contribution

$$\Pi_{sg}^{\text{down, policy}} = -\, B_s\, \kappa_s\, \ln s_{sgh},$$

so downstream surplus varies with policy only through the domestic expenditure share.

Politically Weighted Objective. For each sector–good pair (s, g), the government evaluates tariff policy using a politically weighted objective that combines downstream surplus, domestic upstream rents, foreign upstream rents, and tariff revenue. Let β_s weight downstream surplus, λ_s weight domestic upstream surplus, and γ_{si} weight foreign upstream surplus from origin $i \in \{c, f\}$. The government's per–good objective is

$$\Omega_{sg} = \lambda_s\, \Pi_{sgh}^{\text{up}} \ + \ \beta_s\, \Pi_{sg}^{\text{down, policy}} \ + \ TR_{sg} \ + \sum_{i \in \{c,f\}} \gamma_{si}\, \Pi_{sgi}^{\text{up}}.$$

Under the specific–factor supply structure, upstream rents are a fixed fraction of revenue. Using the downstream expenditure level $M_{sg} = B_s$,

$$\Pi_{sgh}^{\text{up}} = \frac{\omega_s}{1+\omega_s} B_s s_{sgh}, \qquad \Pi_{sgi}^{\text{up}} = \frac{\omega_s}{1+\omega_s} \frac{B_s s_{sgi}}{1+\tau_{sgi}}, \qquad TR_{sg} = \sum_{i \in \{c,f\}} \frac{\tau_{sgi}}{1+\tau_{sgi}} B_s s_{sgi}.$$

Substituting these expressions together with $\Pi_{sg}^{\text{down, policy}} = -B_s \kappa_s \ln s_{sgh}$ and dividing by $M_{sg} = B_s$ yields the normalized per–good objective:

$$\Omega_{sg} = \lambda_s \left(\frac{\omega_s}{1+\omega_s} \right) s_{sgh} - \beta_s\, \kappa_s\, \ln s_{sgh} + \sum_{i \in \{c,f\}} \left(\frac{\tau_{sgi} + \frac{\omega_s}{1+\omega_s} \gamma_{si}}{1+\tau_{sgi}} \right) s_{sgi}, \qquad s_{sgh} = 1 - s_{sgc} - s_{sgf}.$$

Because final-good prices are pinned down in world markets, consumer expenditure does not vary with a tariff exclusion; all variation in Ω_{sg} arises through changes in upstream rents, downstream surplus, and tariff revenue. The economic type in the signaling game is therefore

$$\theta_g \equiv \Omega_{sg}(\tau_{sgc} = 0) - \Omega_{sg}(\tau_{sgc} > 0),$$

the change in the government's politically weighted objective from removing the Chinese tariff. A positive θ_g indicates that granting an exclusion increases the government's objective.

4.9 Equilibrium Implications and Connection to Welfare

The cutoff equilibrium characterized above delivers three equilibrium objects that map directly into the moments used for calibration: (i) the share of firms that lobby, determined by the participation cutoff $\underline{\theta}$; (ii) approval rates conditional on lobbying status, P_1 and P_0; and (iii) the optimal Section 301 tariff implied by the government's politically weighted objective.

These objects depend on two sets of primitives. The informational primitives $(c, \sigma_\varepsilon, \eta)$ govern lobbying incentives and the informativeness of signals, while the sector-specific structural primitives—including technology parameters $(B_s, \sigma_s, \omega_s)$ and political weights $(\lambda_s, \beta_s, \gamma_{si})$—shape the underlying economic impacts θ_g and therefore the model-implied optimal tariff. Together, these primitives determine the equilibrium mapping from economic fundamentals to lobbying behavior, approval patterns, and tariff policy.

In the model, informational lobbying improves targeting conditional on participation, since higher types generate larger gains under the government's objective. At the same time, lobbying also alters the tariff that the government chooses ex ante and changes how it interprets the absence of lobbying in equilibrium. These forces, together with the cost of participation, imply that the net effect of lobbying on welfare is theoretically ambiguous. The quantitative analysis evaluates the overall contribution of the informational channel in light of these competing forces.

5 Quantitative Implementation and Calibration

5.1 Calibration Environment and Parameter Structure

The quantitative implementation combines a small set of economy-wide signaling parameters with sector-specific heterogeneity to match the observed cross-sector variation in tariff levels, lobbying participation, and approval outcomes. Three parameters are common across all sectors: the fixed participation cost of lobbying c, the standard deviation of the noisy signal σ_ε, and the standard deviation of the idiosyncratic political shock η. These parameters jointly determine the informativeness of lobbying, the randomness in approval decisions, and the overall incidence of participation in equilibrium.

Each sector s has two additional parameters. The dispersion parameter α_{0s} governs the heterogeneity of product-level sourcing patterns within the sector, and therefore the dispersion of the welfare-relevant types $\{\theta_g\}$. The political weight γ_{cs} determines how the policymaker values Chinese upstream rents when setting the Section 301 tariff. The weight on non-Chinese foreign upstream rents, γ_{fs}, is set to zero, reflecting that the Section 301 investigation focused exclusively on Chinese suppliers.

A central feature of the environment is that the Section 301 tariff is chosen in a Stage 1 problem that anticipates equilibrium behavior in the exclusion stage. Given the sector's economic fundamentals (σ_s, ω_s) and its political weights, the government selects the tariff that maximizes its politically weighted objective while accounting for how the distribution of types influences firms' lobbying decisions and exclusion outcomes. This forward-looking mapping yields, for each sector, a model-implied Section 301 tariff, the distribution of $\{\theta_g\}$, the lobbying share, and the approval probabilities.

The political weights on domestic upstream rents and downstream surplus, (λ_s, β_s), are pinned down analytically from the sector's MFN (Column 1) and Column 2 tariffs, following Chor et al. (2025). Once these two components are fixed, the only remaining political-economy force shifting the optimal Section 301 tariff is γ_{cs}. The pair $(\alpha_{0s}, \gamma_{cs})$ therefore jointly determines the dispersion and level of the type distribution, and through them the sectoral patterns of lobbying and approval.

Given any candidate values of $(\alpha_{0s}, \gamma_{cs})$ and the common parameters $(c, \sigma_\varepsilon, \eta)$, the model simulates the implied type distribution, solves the signaling-game equilibrium, and computes four sectoral moments: (i) the optimal Section 301 tariff, (ii) the lobbying share, (iii) the approval rate among lobbyists, and (iv) the approval rate among non-lobbyists. These moments discipline the common parameters in the outer loop through their variation across sectors. Once $(c, \sigma_\varepsilon, \eta)$ are fixed, each sector has exactly one parameter, γ_{cs}, that shifts the tariff level and one parameter, α_{0s}, that governs dispersion. Because the three signaling-related moments arise from a single cutoff rule that separates lobbyists from non-lobbyists, sectors generally cannot match all moments exactly. The resulting residual reflects the structural restrictions of the model.

The common parameters have clear interpretations. The cost c governs the extensive margin of lobbying; a higher cost discourages participation and compresses the share of lobbying firms. The signal noise σ_ε determines how informative lobbying is about the firm's type. The political shock η governs residual randomness in approval outcomes. The next subsection describes how these ingredients enter a nested calibration procedure that aligns the model with the cross-sectoral patterns in the data.

5.2 Heuristic Identification

Although the mapping from parameters to moments is fully equilibrium-based, it is helpful to describe heuristically how cross-sectoral variation disciplines the three common signaling parameters. The central idea is that the four sectoral moments respond differentially to changes in the cost of lobbying, the informativeness of the signal, and the magnitude of the political shock.

Lobbying participation and the cost of lobbying. Holding the informational environment fixed, the fixed cost c determines the cutoff type in the signaling game and therefore governs the share of firms that choose to lobby. Higher values of c raise the cutoff and reduce participation. Changes in σ_ε and η affect participation only indirectly through expected approval probabilities. As a result, cross-sectoral variation in lobbying shares $\{L_s\}$ primarily disciplines the level of c.

Approval gaps and signal informativeness. The difference in approval rates between lobbyists and non-lobbyists, $A_{1s} - A_{0s}$, is informative about the precision of the lobbying signal. In the model, the signal noise σ_ε directly affects only the approval probability of lobbyists, because non-lobbyists generate no signal and are evaluated using the truncated prior over types below the cutoff. As σ_ε changes, the government's posterior for lobbyists places different weight on realized signals relative to the lobbying prior, and the equilibrium cutoff adjusts, altering the composition and approval rate of the lobbying pool. In contrast, the political shock η primarily shifts approval probabilities for both lobbyists and non-lobbyists in similar ways, so it has a more limited effect on the *difference* between their approval rates. Sectoral variation in $A_{1s} - A_{0s}$, taken together with observed dispersion in outcomes and lobbying shares, therefore provides useful discipline for the signal noise parameter σ_ε.

Approval levels and political noise. The political shock η directly governs the residual randomness in approval decisions for both lobbyists and non-lobbyists. Larger values of η widen the support of the approval decision around the deterministic threshold and therefore raise approval among non-lobbyists in particular, since many non-lobbyists are evaluated close to the cutoff. In contrast, σ_ε primarily influences the sorting of firms into lobbying and affects non-lobbyist approval only indirectly through the equilibrium cutoff. For this reason, the model relies on the level of non-lobbyist approval rates $\{A_{0s}\}$—which are especially sensitive to η—to discipline the magnitude of the political shock.

Taken together, these heuristic patterns show how the three approval- and participation-based moments jointly inform the three common signaling parameters. No single moment isolates one parameter on its own, but the cross-sectoral variation in lobbying participation, approval gaps, and approval levels provides enough joint discipline to pin down the cost of lobbying, the informativeness of the signal, and the magnitude of political noise that characterize the signaling environment.

5.3 Calibration Structure

The calibration exploits four empirical moments for each sector s: (i) the observed Section 301 tariff $\tau_{s,301}$; (ii) the share of firms that lobbied L_s; (iii) the approval rate among lobbyists A_{1s}; and (iv) the approval rate among non-lobbyists A_{0s}. Taken together, these moments summarize how participation and exclusion outcomes vary across sectors, conditional on observed trade structure and political weights determined outside the calibration.

The nested calibration consists of three layers:

1. **Inner layer.** For given $(\alpha_{0s}, \gamma_{cs})$ and $(c, \sigma_\varepsilon, \eta)$, the model constructs the distribution of types $\{\theta_g\}$, solves the signaling-game equilibrium, and produces the implied values of $(\tau_{s,301}, L_s, A_{1s}, A_{0s})$.
2. **Sector-level profiling.** Holding the common parameters fixed, each sector chooses $(\alpha_{0s}, \gamma_{cs})$ to minimize the squared deviations between the four model-implied moments and their empirical counterparts. The minimized value defines the sector-specific loss $\ell_s(c, \sigma_\varepsilon, \eta)$.
3. **Outer loop.** The common parameters $(c, \sigma_\varepsilon, \eta)$ are chosen to minimize the aggregate loss $\sum_s \ell_s(c, \sigma_\varepsilon, \eta)$.

This structure isolates the informational primitives at the economy level while allowing each sector to flexibly match its tariff and exclusion patterns through its own heterogeneity and political-weight parameters.

5.4 Sector-Level Profiling

Sector-level heterogeneity is captured through the two parameters $(\alpha_{0s}, \gamma_{cs})$, which shape how exclusion decisions respond to the sector's fundamentals.

Dispersion of product-level impacts. Product-level sourcing shares (s_{gh}, s_{gc}, s_{gf}) are drawn from a Dirichlet distribution centered on the sector's observed pre–trade-war sourcing pattern, with total mass α_{0s}. Lower values of α_{0s} generate greater dispersion, producing a wider distribution of welfare impacts $\{\theta_g\}$ and more cross-sectional variation in lobbying incentives. Higher values compress the type distribution around the sector mean.

Political sensitivity to Chinese supply. The parameter γ_{cs} governs the weight placed on Chinese upstream rents and therefore shifts the optimal Section 301 tariff. The weights on domestic upstream rents and downstream surplus, (λ_s, β_s), are determined from MFN and Column 2 tariffs, while the weight on non-Chinese foreign upstream rents is set to zero. Conditional on (σ_s, ω_s), the pair $(\alpha_{0s}, \gamma_{cs})$ therefore jointly determines the sector's implied distribution of types and its Section 301 tariff.

Profiling procedure. For each candidate $(c, \sigma_\varepsilon, \eta)$, the model computes the sector's four implied moments for every trial pair $(\alpha_{0s}, \gamma_{cs})$. The profiling step chooses the pair minimizing the loss

$$(\alpha_{0s}, \gamma_{cs}) = \arg\min_{(\alpha,\gamma_c)} \Big[(\tau_{s,301} - \tau_{s,301}^{\text{model}})^2 + (L_s - L_s^{\text{model}})^2 + (A_{1s} - A_{1s}^{\text{model}})^2 + (A_{0s} - A_{0s}^{\text{model}})^2\Big].$$

Because (λ_s, β_s) are pinned down by observed MFN and Column 2 tariffs, each sector's remaining heterogeneity is summarized by the pair $(\alpha_{0s}, \gamma_{cs})$. These two parameters jointly determine the model-implied tariff and the three signaling-related moments. However, the signaling moments (L_s, A_{1s}, A_{0s}) all arise from a single cutoff equilibrium, so the sector cannot generally match all of them exactly. In practice, the fit for A_{0s} in particular reflects this structural restriction.

5.5 Calibrated Results

Table 4: Calibrated Common Parameters and Economic Interpretation

Parameter	Value	Economic Meaning
c	0.00020	Lobbying cost per dollar of expenditure; higher $c \Rightarrow$ fewer firms lobby.
σ_ε	9.7×10^{-7}	Std. dev. of signal noise; small $\Rightarrow$ highly informative signals.
η	0.00070	Std. dev. of political shock; moderate relative to within-sector θ dispersion.

Table 4 reports the estimated common parameters. All parameter values are expressed per dollar of total sectoral expenditure, including both domestic production and imports.

The calibrated signal noise is small ($\sigma_\varepsilon^* = 9.7 \times 10^{-7}$), implying that lobbying conveys highly precise information about firms' types. The calibrated political shock is modest: $\eta^* = 7.0 \times 10^{-4}$ is about forty percent of the median within-sector standard deviation of θ (1.7×10^{-3}). Approval decisions therefore exhibit some randomness but remain highly predictable on average. The primary friction in the signaling environment is instead the participation cost, $c^* = 2.0 \times 10^{-4}$.

Table 5 summarizes the exogenous trade-structure inputs, which anchor the tariff component of the calibration. Table 6 compares the model-implied moments to their empirical counterparts. The model matches the average Section 301 tariff as well as the extensive and intensive margins of lobbying. The approval rate among non-lobbyists is fitted less tightly. Because each sector has only two parameters to fit four moments, the model cannot in general match all moments exactly. In practice, the approval rate among non-lobbyists, A_{0s}, tends to exhibit the largest residual, reflecting the fact that it depends on the entire lower portion of the type distribution induced by the cutoff rule.

Table 5: Pre-Determined Trade Structure (Exogenous Inputs to Calibration)

Measure	Value
Chinese import share	0.067
Rest-of-world share	0.171
Column 1 (MFN) tariff	0.032
Column 2 tariff	0.380

Table 6: Fitted Moments: Lobbying and Approval Outcomes

Measure	Data Mean	Model Mean
Section 301 tariff (additional)	0.204	0.207
Lobbying share	0.218	0.207
Lobbyist approval rate	0.248	0.231
Non-lobbyist approval rate	0.094	0.031

To assess the informativeness of signals, I compare the calibrated noise level to the dispersion of θ across goods within each sector. Across sectors, the median within-sector standard deviation is 1.7×10^{-3}, implying that the calibrated signal noise is several orders of magnitude smaller than underlying heterogeneity. Even in the least dispersed sector, the noise level remains below one percent of the natural variation in θ. Lobbying therefore transmits highly precise information about the welfare consequences of tariff relief.

To interpret the magnitude of c^*, consider a representative sector with \$100 million in total expenditure (including domestic production and imports), of which roughly \$6 million originates from China and faces an additional tariff burden of about $1 million. For interpretive simplicity, I abstract from endogenous responses within the sector, such as tariff pass-through, changes in sourcing shares, and the resulting adjustments in the distribution of welfare impacts $\{\theta_g\}$. These forces matter for the full equilibrium analysis in Section 6, but are second-order for the narrower task of translating the calibrated cost parameter into a dollar value. Under this benchmark, a lobbying cost of $c^* = 2 \times 10^{-4}$ corresponds to an outlay of roughly $20,000. This level is consistent with the fixed-cost nature of filing, legal assistance, and documentation. Aggregating across sectors, the model implies a total lobbying cost of $0.037 billion, comparable in magnitude to aggregate expenditures recorded in lobbying disclosures for Section 301 issues.

5.6 Discussion

The calibrated model reproduces the overall magnitude of lobbying and approval behavior under Section 301. A central conclusion that emerges from the calibration is that the informational channel was not constrained by noisy signals. The estimated noise level σ_ε^* is small relative to the natural dispersion of welfare impacts within sectors, indicating that once firms chose to lobby, their submissions conveyed highly informative signals about the consequences of tariff relief. This pattern is consistent with the institutional setting: exclusion petitions were prepared by professional legal counsel, supported by detailed documentation, and evaluated by a technically trained bureaucracy. When information was supplied, it was clear.

The binding friction instead arises from the fixed cost of participation. The calibrated cost c^* implies that only firms with sufficiently large expected gains from an exclusion found it worthwhile to prepare and submit petitions. In equilibrium, lobbying therefore operates as a selective communication channel: signals are highly precise, but only a subset of firms incurs the cost to transmit them. The political shock η^* introduces some randomness into approval decisions, but is modest relative to underlying heterogeneity, implying that the agency's decisions were reasonably predictable once information was available.

Taken together, these findings suggest that the Section 301 exclusion process functioned primarily as a mechanism for aggregating high-quality information under a politically weighted objective. The informational friction did not lie in the government's ability to interpret signals, but in the cost borne by firms to generate and submit them. This distinction is central for the counterfactual analysis that follows, where the welfare consequences of removing lobbying arise not because lobbying is noisy, but because the government loses access to precise information supplied by participating firms.

6 Counterfactual Analysis

This section quantifies how informational lobbying affects equilibrium outcomes by comparing the calibrated economy with a series of counterfactuals that vary the availability or timing of lobbying. All comparisons are expressed relative to the cooperative MFN benchmark.[5]

6.1 Baseline Welfare Comparison

Table 7 reports the aggregate comparison between the calibrated equilibrium with lobbying and a counterfactual in which lobbying is ruled out entirely. The equilibrium with lobbying generates a welfare cost of $2.12 billion, compared with $2.59 billion when lobbying is absent. Informational lobbying therefore reduces the welfare loss by roughly 18 percent. This reduction reflects improved targeting of tariff relief, since the policymaker can draw on firm-supplied information when deciding which products to exempt. Because the political weights remain unchanged across scenarios, the gains arise from better information rather than altered political preferences. Importantly, the informational benefits outweigh the participation costs, yielding a net welfare improvement.

Table 7: Aggregate Welfare Comparison: With vs. Without Lobbying (Relative to MFN)

Measure	With Lobbying ($bn)	No Lobbying ($bn)	Ratio (With/No)
Total Welfare Cost	2.118	2.593	0.82

6.2 Mechanisms and Counterfactual Scenarios

In addition to the baseline comparison between the equilibria with and without lobbying, we consider two counterfactual exercises that vary the timing or availability of lobbying while holding all structural parameters fixed at their calibrated values. These scenarios isolate the mechanisms through which information affects outcomes by comparing economies in which the policy rule is held fixed but participation and beliefs change ex post.

1. **CF1: Surprise Shutdown.** Lobbying ceases after the tariff rule has already been set under the expectation of information.[6]

2. **CF2: Surprise Lobbying.** Lobbying becomes available after the tariff was chosen for a world without information.[7]

[5]Throughout this section, positive values denote welfare *losses* relative to MFN.

[6]In CF1, the *policy rule* is fixed at the equilibrium-with-lobbying tariff, but participation and signals are switched off ex post. Only participation, beliefs, and approval probabilities adjust.

[7]In CF2, the *policy rule* is fixed at the no-lobbying tariff, but the signaling game is turned on ex post.

Figure 1 summarizes the structure of the four scenarios.

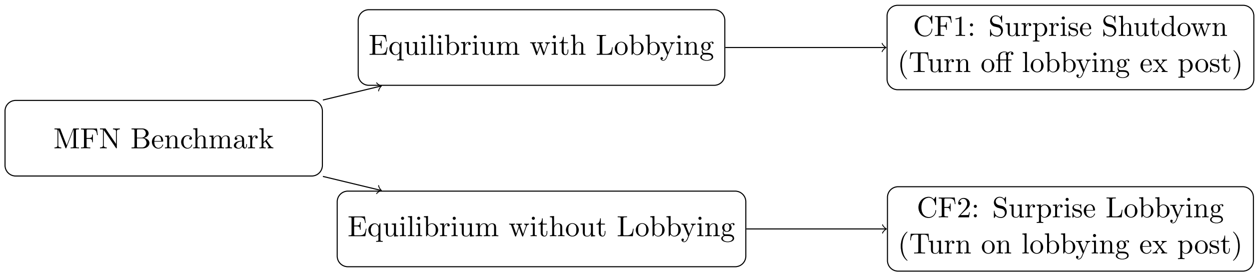

Figure 1: Structure of Counterfactual Scenarios

6.3 Tariffs and Approval Rates Across Scenarios

Informational lobbying affects both the *tariff rule* chosen ex ante and the *approval probabilities* applied ex post.

Tariffs. In the equilibrium with lobbying, the endogenous average Section 301 tariff is $\tau_{301}^{*} = 0.207$. In the no-lobbying equilibrium, the average tariff rises to $\tau_{301}^{\mathrm{NL}} = 0.243$. The difference reflects a simple force: when lobbying is available, a higher tariff induces more petitions and thus higher participation costs, which weakens the policymaker's incentive to set the tariff aggressively. When lobbying is absent, this deterrent disappears and the optimal tariff is correspondingly higher.

Approval rates. With lobbying, the average approval probability is 7.88 percent. In the no-lobbying equilibrium, where the policymaker observes no informative signals, the average approval probability falls to 4.83 percent. CF1 and CF2 lie between these extremes, with approval probabilities moving in the direction implied by the presence or absence of informative signals while the tariff rule is held fixed.

Table 8: Average Tariffs and Approval Probabilities Across Scenarios

Scenario	Avg. Tariff (additional) τ_{301}	Avg. Approval Prob. (%)
Equilibrium with Lobbying	0.207	7.88
Equilibrium without Lobbying	0.243	4.83
CF1: Surprise Shutdown	0.207	5.31
CF2: Surprise Lobbying	0.243	7.45

Participation, beliefs, and approval probabilities adjust accordingly.

6.4 Welfare Decomposition Across Scenarios

To clarify the mechanisms behind the aggregate results above, Table 9 reports how each component of the government's politically weighted objective responds to the counterfactual variation in lobbying. Panel A decomposes the changes into upstream producer surplus, downstream producer profits, tariff revenue, and lobbying costs. Panel B reports the implied total welfare loss in each scenario, allowing a direct comparison of the equilibrium with lobbying, the equilibrium without lobbying, and the two mixed-timing environments (CF1 and CF2).

Table 9: Welfare Decomposition Across Counterfactuals (Relative to MFN)

Panel A: Components of Welfare Change	**Δ relative to MFN ($bn)**			
Scenario	Prod. Surplus	Downstream Profit.	Tariff Rev.	Lobby Cost
Equil. with Lobbying (Baseline)	−0.46	7.99	−5.45	0.04
Equil. no Lobbying	−0.49	9.24	−6.16	0.00
CF1: Surprise Shutdown	−0.47	8.37	−5.73	0.00
CF2: Surprise Lobbying	−0.48	9.04	−6.01	0.05

Panel B: Aggregate Welfare Cost	**Total ($bn)**	**Ratio (to Baseline)**
Equil. with Lobbying (Baseline)	2.12	1.00
Equil. no Lobbying	2.59	1.23
CF1: Surprise Shutdown	2.17	1.02
CF2: Surprise Lobbying	2.60	1.23

6.5 Interpretation

The counterfactuals separate two channels: (a) the impact of information when the *tariff rule* is held fixed, and (b) the impact of the *tariff rule itself* when chosen under different information expectations.

(i) Fixed–policy comparisons. Holding the tariff rule fixed but turning lobbying off (CF1) raises the total welfare loss only slightly, from $2.12 to $2.17 billion. Turning lobbying on after the no-lobbying tariff rule has been chosen (CF2) likewise generates only a small change, from $2.59 to $2.60 billion. These comparisons show that participation, signals, and lobbying costs have largely redistributive effects under a given tariff rule; they are not a first-order source of aggregate welfare differences.

(ii) Policy-rule effect. The dominant quantitative effect comes from the choice of the tariff rule. As shown in Table 8, the government sets a substantially lower tariff when it anticipates lobbying (0.207 versus 0.243). This moderation reflects the mechanism that a higher tariff induces more firms to incur the fixed cost of petitioning, which the policymaker internalizes when choosing the rule.

Information then improves ex-post implementation. With lobbying, the approval rate is 7.9 percent rather than 4.8 percent, reflecting more precise identification of high-harm products. The combination of a more moderate initial tariff and more targeted relief accounts for nearly the entire $470 million difference in aggregate welfare costs between the two equilibria.

(iii) Composition. Differences in downstream profits explain most of the welfare gap. The no-lobbying equilibrium combines the worst of both worlds: a higher, blunt tariff that depresses downstream profits and an inability to screen, which leaves high-harm products unrelieved. The baseline equilibrium mitigates both distortions.

(iv) Summary. The welfare value of informational lobbying does not come from its direct, redistributive effects under a fixed tariff rule, but from the existence of the channel itself. Its availability disciplines the policymaker to choose a more moderate tariff ex ante and enables more efficient, better-targeted exclusions ex post.

7 Conclusion

This paper develops and quantifies a model of *informational lobbying* in which firms lobby not to purchase political access but to transmit costly information to a bureaucracy operating under political constraints. Using the U.S.–China Section 301 tariff exclusion process as a case study, the analysis shows that lobbying can improve the quality of policy implementation even when the policymaker's objective is explicitly political.

The model embeds a noisy signaling game inside a politically weighted objective framework. Firms observe the product-level impact of tariff relief under the government's own objective and decide whether to incur a cost to reveal that information. The government, in turn, interprets lobbying behavior through Bayesian updating and decides on exclusions subject to political shocks. When signals are precise but participation is costly, lobbying endogenously selects firms whose information is most valuable to the policymaker, generating targeted relief without altering the political weights that shape the underlying tariff.

Calibrated to the universe of 50,000 petitions, the model matches key patterns of lobbying participation and approval rates across sectors. The estimated lobbying cost is modest but sufficient to deter many firms from engaging, while the inferred signal noise is near zero and political shocks are modest. These estimates indicate that most uncertainty in the exclusion process arose from limited participation rather than from arbitrariness.

The quantitative results reveal a conceptually simple but powerful mechanism. Informational lobbying lowers the average Section 301 tariff from 0.243 to 0.207 because the policymaker internalizes that a higher tariff induces more firms to incur the fixed cost of petitioning. Once the tariff is chosen, lobbying provides precise signals that raise the approval rate from 4.8 to 7.9 percent, enabling more targeted relief for the most adversely affected products. Together, tariff moderation and better targeting reduce the aggregate welfare cost from $2.59 to $2.12 billion—an 18 percent improvement relative to a world without lobbying.

The counterfactuals clarify that these gains arise from the *existence* of the informational channel rather than from lobbying activity per se. Turning lobbying on or off after the tariff rule has been set affects welfare only mildly: participation costs are second-order, and information is valuable only when the policymaker anticipates it. The main effect comes from how the policymaker chooses the tariff in anticipation of the informational environment. This interaction between discretion and information generalizes to other regulatory settings in which agencies implement politically directed policies using stakeholder-provided information.

Implications and Directions for Future Research

The findings illustrate how bureaucratic discretion can, under certain conditions, harness rather than amplify political distortion. In trade policy, this helps explain why administrative exemption procedures such as product-level rulings, waiver systems, and licensing reviews persist even in highly politicized contexts: they enable policymakers to incorporate decentralized information while retaining political control.

Several extensions follow naturally. First, firms in the model act independently; introducing strategic interaction could illuminate how information aggregates within sectors and how free-riding affects participation. Second, the calibration treats political weights as reduced-form objects; linking them to industry characteristics or electoral geography could connect informational lobbying to the broader determinants of protection. Third, allowing for heterogeneous participation costs across firms would generate richer sorting patterns and could reveal which types of firms are most constrained by the fixed-cost nature of petitioning.

More broadly, the framework offers a template for quantifying the informational content of lobbying in other administrative environments. Policy regimes such as Buy America waivers, export controls, foreign-investment screening, and environmental permitting share the same institutional logic: policymakers must implement politically directed rules using private information held by firms. Applying similar signaling structures to these settings could deliver a unified view of how information flows shape the effectiveness and legitimacy of modern industrial and trade policy.

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